Section-17

TECHNICAL SPECIFICATION FOR HOT DIP GALVANISED BOLTS & NUTS AND WASHERS FOR TRANSMISSION LINE TOWERS

1.0 SCOPE

This part of the specification provides for procurement, inspection, stage testing and testing before despatch, packing and delivery FOR destination of Galvanized Bolts & Nuts, Electro Galvanised Rectangular Spring Washers and Flat Washers for use on EHT transmission towers in various lines and Sub-stations of the State. These bolts & nuts and washers are to be used for single, double and Multi circuit 132kV, 220kV transmission line towers and for structures in 132kV & 220 kV substations of the owner.

2.0 It is not the intent to specify completely herein all the details of the design and manufacturing of GI Bolts & Nuts and washers. However, the material shall conform in all respects to high standards of Engineering, Design and workmanship and shall be capable of performing in continuous commercial operation. All such things shall be deemed to be within the scope of the Bidders supply irrespective of whether those are specifically brought out in this specification and/or commercial order or not.

3.0 STANDARDS :

The Bolts, Nuts and Washers shall conform to the following Indian Standards, which shall mean latest revisions, amendments/changes adopted and published, unless otherwise specified herein before. Indian standards to which these standards generally correspond are also listed below:

Sl. No.	IS No.	TITLE
1.	IS-12427	Specification for Transmission Tower Bolts
2.	IS-1367	Technical Supply Conditions for Threaded Steel Fasteners
3.	IS-1367 (Part-2)	Product grades and tolerances
4.	IS-1367 (Part-3)	Mechanical properties and test methods for bolts, screws and studs with full load ability.
5.	IS-1367 (Part-6)	Mechanical properties and test methods for nuts with specified proof loads.
6.	IS-1367 (Part-13)	Hot dip galvanised coatings on threaded fasteners.
7.	IS-1363 (Part-3)	Hexagonal head bolts, screws and nuts of product grade 'C' – Hexagon nuts (size range M-5 to M-64).
8.	IS-2633	Methods for testing uniformity of zinc coated articles.
9.	IS-3063	Single coil rectangular section spring lock washers
10.	IS-4072	High quality spring steel
11.	IS-1573	Electro galvanization
12.	IS-2016	Plain washers
13.	IS-6821	Testing of washers
14.	IS-2629	Recommended practice for Hot dip galvanising of iron and steel
15.	IS-4759	Hot dip zinc coatings on structural steel and other allied products

16.	IS-6745	Determination of Mass of Zinc coating on zinc coated iron and steel articles
17.	IS-2633	Methods for testing uniformity of coating on zinc coated articles
18.	IS-5369	General requirements for plain washers and lock washers
19.	IS-2614	Methods for sampling of fasteners.

4.0 PRINCIPAL PARAMETERS
DIMENSIONS

4.1 BOLTS :

4.1.1 The dimensions of the Bolts shall be as given in table-I of IS-12427.

4.1.2 Preferred length size combinations as well as Grip ranges shall be as given in table - II of IS:12427.

4.1.3 Prior to Hot-dip galvanising, the bolt threads shall conform to tolerance class-8 (G) of IS-4218 (Part-6)-1978 and the latest revision if any.

4.2 Nuts :

The dimensions of the standard nuts shall be as given in Table-I of IS-1363 (Part-3) - 1992 :

4.3 Spring Washers :

The dimensions of the spring washers shall be as given in Table-1A of IS-3063-1994.

4.4 Flat Washers:

4.4.1 Flat Washers shall be of 40 mm. Dia. Round of thicknesses 4 mm, 5 mm, 6 mm, 8 mm, 10 mm and 12 mm suitable for 16 mm dia. Bolts for use in transmission line towers.

4.5 DRAWINGS

The Bidder shall furnish the guaranteed technical particulars of the materials offered. Wherever necessary dimensioned drawings shall be furnished along with the Bid.

5.0 GENERAL TECHNICAL REQUIREMENTS:

5.1 Hexagon Head Bolts

5.1.1 The bolts shall conform to IS:12427-1988 and its latest revision if any .

5.1.2 The bolts shall be of product grade 'C' as specified in IS:1367 (Part-2)-1979.

5.1.3 The bolt shall be of property class 5.6 and 8.8 as specified in IS:1367 (Part-3)-1991 (latest revision)

5.2 Hexagon nuts :

5.2.1 The nuts to be used for the above bolts shall conform to the requirements of IS:1363 (Part-3) -1992. The nuts shall be of property class 5 as specified in IS:1367 (Part-6)-1980. The proof Stress Value shall be as follows:

Nominal Nut Size	**Proof Stress (SP) N/Sq.mm**
M 16	500
M 24	500

5.2.2 The bolts & nuts and Washers shall be hot dip galvanised in accordance with the requirements of IS:1367 (Part-13)-1991(Latest revision) and tested for uniformity of coating as per IS:2633.

5.3 Spring Washers :

5.3.1 The spring washers shall be made from high quality spring steel conforming to IS-4072-1975, including its latest revision if any

5.3.2 The spring washers shall be of type B and shall confirm to IS-3063-1994, including latest revision if any. Type B spring lock washers with flat (Squares) ends

5.3.3 The spring washers shall be electro galvanized conforming to Grade 4 IS-1573-1986 with a coating thickness of minimum 25 microns (local thickness)

5.4 Flat washers :

5.4.1 The flat washers shall be 40mm round made from mild steel with thickness of 4mm, 5mm, 6mm, 8mm, 10mm and 12mm suitable for 16mm dia bolts for Transmission line towers

5.4.2 The flat washers shall be hot dip galvanized conforming to IS-2629-1985 including latest revision if any.

5.5 MECHANICAL PROPERTIES

Shall conform to IS-1367 (Part-3)-1991 for Bolts.
IS-1367 (Part-6)-1980 for Nuts
IS-3063-1994 for Spring washers
IS-2016-1967, IS-6821-1973, IS-1367(part-3)-1991 for Flat washers

5.6 DESIGNATION

5.6.1 The Bolts shall be designated as per IS-12427-1988 or latest revisions if any.

Transmission tower bolt M 16 x 50 N IS - 12427.

The Nuts shall be designated as per IS-1363 (Part III)-1992 or latest revisions if any.

Hexagon nut M 16.5 - IS 1363 (Part-3).

The spring washers shall be designated as per IS-3063-1994 or latest revisions if any.

Spring lock washer B 16 -IS 3063 Electro galvanised.

The Flat washers shall be designated as per IS-2016-1967 and 6821-1973 or latest revisions if any.

Punched washer A 16 IS - 2016 steel.

5.7 SAMPLING

5.7.1 Shall be in accordance with the IS-2614-1969 and IS-6821-1973, its latest revisions if any for Bolts & Nuts and Washers respectively.

5.8 MARKING

5.8.1 Tower Bolts shall be marked with the following symbols on the top surface of the bolt-head either embossed or indented as given below:

a) The manufacturers identification symbol and

b) Transmission tower bolt identification symbol 'T'.

5.8.2 The hexagon nuts shall be marked and delivered in accordance with IS:1367 (Part-18) - 1979 marking and mode of delivery.

5.8.3 The packages containing washers shall be marked with the size and manufacturer’s name or trade mark.

5.9 FINISH

The Bolts, Nuts and Flat washers shall be hot dip galvanised in accordance with the requirements of IS-1367 (Part-13)-1983,IS-2633-1986 and IS-4759-1984 or latest revisions if any. The Spring washers shall be electro galvanized in accordance with the requirements of IS-1573-1986 or latest revision if any.

6.0 TESTS :

The sampling of fasteners shall be as per IS-2614 for ensuring visual and dimensional characteristics and physical properties. The bolts shall be tested as per IS-1367 (Part-3)-1991 for mechanical properties. The nuts shall be tested as per IS-1367 (Part-6)-1992 for mechanical properties.

6.1.1 For tensile, proof load and wedge-loading tests three threads only shall be exposed between the grips. This is obtained by freely running the nut or fixture to the fullest extent and then unscrewing the specimen three full turns. These tests are to be done after chemically degalvanizing the bolts.

6.1.2 Bolts shall meet the hardness requirements (clause 5.1.2. of IS-12427-1988)

6.1.3 The Acceptance and Routine tests shall be conducted on Bolts as per IS 1367 (Part 3) -1991.

6.2 The Acceptance and Routine tests shall be conducted on Hexagon Nuts as per IS-1367 (Part-6) -1992.

6.3 The Acceptance and Routine tests shall be conducted on Spring Washers as per IS 3063-1994 and Flat Washers as per latest editions of IS 2016-1967 and IS 6821-1973.

6.4 <u>Test Facilities</u>

6.4.1 The supplier must have testing facilities for all the tests mentioned or the test should be carried out through a reputed/Government undertaking after taking the approval of Purchaser. The expenditure towards carrying out the entire test is to the account of the supplier.

6.4.2 Copies of Acceptance test reports shall be furnished in at least six (6) copies along with one original. One copy shall be returned duly certified by the Purchaser only after which the material will be dispatched.

6.4.3 Record of routine test reports shall be maintained by the supplier at his works for periodic inspection by the Purchaser's representative.

6.5 Test Certificates of test conducted during manufacture shall be maintained by the supplier. These shall be produced for verification as and when desired by the Purchaser.

6.6 <u>Test Procedures</u>: As detailed in Annexure enclosed.

7.0 INSPECTION

7.1 The Purchaser's representative shall, at all times, be entitled to have access to the works and all places of manufacture where bolts, nuts, spring washers and flat washers shall be manufactured and the representative shall have full facilities for unrestricted inspection of the supplier's works, raw materials and process of manufacture and conducting necessary tests as detailed herein.

7.2 The supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of bolts and nuts in its various stages so that arrangements can be made for inspection.

7.3 Fasteners shall be selected and examined for each lot separately for ascertaining their conformity to the requirements of the relevant specification.

7.4 The number of fasteners to be selected from a lot shall depend upon the size of the lot and shall be in accordance with column 1 to 3 of Table-1 of latest revisions of IS-2614 for normal inspection. Inspection of physical properties is as per column 1 and 2 of Table-2 of IS-2614.

7.5 A lot shall satisfy the requirements as per clauses 4.1.1 & 4.1.2, 4.2 and 4.3 of IS-2614 for the visual inspection, dimensional characteristics and physical properties respectively. In case the number of defectives lies in between the acceptance and rejection numbers, a second sample of same size shall be taken and inspected for the above requirements. The number of defective fasteners found in the first and second samples shall be combined and if the combined number of defects is less than or equal to the corresponding acceptance number, the lot shall be declared as conforming to the requirements for visual inspection, dimensional and physical properties, otherwise not.

7.6 No material shall be dispatched from its point of manufacture before it has been satisfactorily inspected and tested, unless the inspection is waived off by the Purchaser in writing. In the later case also, the bolts, nuts, spring washers and flat washers shall be dispatched only after satisfactory testing for all tests specified herein has been completed and approved by the Purchaser.

7.7 The acceptance of any quantity of material shall in no way relieve the supplier of any of his responsibilities for meeting all requirements of the specification and shall not prevent subsequent rejection if such material is later found to be defective.

7.8 The supplier shall give 15 days (for local supplies)/30 days (for Foreign supplies) advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests.

8.0 DOCUMENTATION

8.1 Six sets of test reports, duly approved by the Purchaser shall be submitted by the supplier for distribution, before commencement of supply. Adequate copies of acceptance and routine test certificates, duly approved by the Purchaser shall accompany the dispatched consignment.

8.2 Approval of drawing/work by Purchaser shall not relieve the supplier of his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices. The material shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of ordering and Purchaser shall have the power to reject any work or material which in his judgment is not in full accordance therewith.

9.0 PACKING & FORWARDING

9.1 The Supplier shall provide such packing of the Goods as is required to prevent their damage or deterioration during transit to their final destination, as indicated in the Contract. The packing shall be sufficient to withstand, without limitation, rough handling during transit and exposure to extreme temperatures, salt and precipitation during transit, and open storage. Packing case size and weights shall take into consideration, where appropriate, the

remoteness of the Goods' final destination and absence of heavy handling facilities at all points in transit.

The packing, marking and documentation within and outside the packages shall comply strictly with such special requirements as shall be explicitly provided for in the Contract, including additional requirements, if any and in any subsequent instructions ordered by the Purchaser.

The correct number of bolts & nuts will be packed in heavy double gunny bags accurately tagged in accordance with the contents. Net weight of each bag shall not exceed 50 Kgs. Each package will be marked with proper paint/indelible ink with the following:

(i) Project: (ii) Contract No.: (iii) country of Origin of Goods:
(iv) Supplier's Name: (v) packing list Reference number.

9.2 The Spring washers and flat washers shall be packed in 50 Kg. Double gunny bags. The material shall be properly packed and dispatched to destinations. Each bag shall contain washers of one size only.

ANNEXURE
TEST PROCEDURES

1.0 **Tests :**

1.1 For tensile, proof load and wedge-loading tests three threads only shall be exposed between the grips.

1.2 Bolts shall meet the hardness requirements.

1.3 **Shear Strength**
Shear strength of the bolts shall meet the requirements given in Table-III

Normal Bolt Thread size	Nominal Shank Area* mm^2.	Minimum Single Shear Load kN +
M 16	201	62
M 24	314	97.3

* based on nominal bolt shank diameter.
+ based on shear stress of 310 MPa.

1.4 **Preece Test** shall be carried out as per IS-1573-1976 and IS-1505-1970 or its latest revisions.

1.5 **Clearing of samples:** Shall be done in accordance with IS-2633-1986.

1.6 **Interpretation of Test:** Shall be carried out in accordance with IS-2633-1986 of its latest revisions if any.

1.7 **Number of Dips:** shall be in accordance with the IS-4826-1979 or its latest revisions if any.

1.8 **Vickers hardness tests:** Shall be carried out in accordance with the IS-1500-1968 or its latest versions.

1.9 **Brinall Hardness Test:** In accordance with the IS-1500-1968 or its latest revisions.

2.0 **Rockwell hardness Test**: Shall be carried out in accordance with the IS-1586-1968 or its latest revisions if any.

2.1 **Proof Load Test:** Shall be carried out in accordance with the IS-1821-1967 or its latest revisions if any.

2.2 **Wedge Load Test** : To be conducted for Bolts & Nuts in accordance with the IS-1367 (Part-3) 1979 or its latest revisions if any.

2.3 **Impart Test:** Shall be carried out in accordance with the IS-1499-1959 or its latest revisions.

2.4 **Head soundness Test**: Shall be carried out in accordance with IS-1367 (Part-3)-1979 or its latest revisions.

2.5 **Surface integrity Test**: Shall be carried out in accordance with the IS-1367 (Part 10)-1979 or its latest revisions.

2.6 **Re-Tempering Test**: In accordance with the IS-1367 (Part 3)-1991 or its latest revisions.

2.7 **Proof Load Test** : This test shall be conducted on nuts in accordance with the IS 1307 (Part-VI) - 1980 or its latest revisions.

2.8 **Hardness Test** : The test shall be conducted on Nuts in accordance with the IS:1367 (Part-VI) - 1980 or its latest revisions.

2.9 **Surface Integrity Test** : On Nuts in accordance with the IS:1367 (Part-X) - 1979 or its latest revisions.

2.10 **Hardness Test** : The hardness of washers shall be tested in accordance with IS:1501 (Part-1) - 1984.

2.11 **Permanent Set Test** } On washers shall be conducted in accordance
} with 3063-1994 and 2016-1967 or latest its revisions.

2.12 **Permanent Load Test** }
}

2.13 **Twist Test** }

GUARANTEED TECHNICAL PARTICULARS FOR G.I. BOLTS & NUTS
(to be filled by the Bidder)

1.	Specifications & Standards	:	
	i) Bolts	:	
	ii) Nuts	:	
	iii) Spring Washers	:	
	iv) Flat washers	:	
2.	Tolerances	:	
3.	Raw material	:	
	i) For bolts, nuts, spring washers & flat washers	:	
	ii) Grade	:	
	iii) Type of steel used	:	
4.	Chemical Composition	:	
	i) For Bolts nuts, spring washers & flat washers	:	
	Carbon (Max.)	:	
	Phosphorus (Max.)	:	
	Sulphur (Max.)	:	
5.	Mechanical Properties	:	
a)	For Bolts	:	
	i) Tensile Strength (N/Sq.mm - min)	:	
	ii) Yield Strength (N/Sq.mm - Min.)	:	
	iii) Stress under proof load (N/Sq.mm – Min.)	:	
	iv) Hardness HRB (Min. / Max.)	:	
	v) Elongation after free - ture (% Min.)	:	
	vi) Strength under wedge load	:	
	vii) Head soundness test	:	
b)	For Hex. Nuts	:	
	i) Proof stress (N/Sq.mm – Min	:	
	ii) Test Hardness	:	
	iii) Surface Integrity test	:	
c)	For Spring washers & flat washers	:	
	i) Hardness test	:	
	ii) Permanent set test	:	
	iii) Permanent load test	:	
	iv) Twist test	:	

6.	Galvanising (Hot Dip – Mass & Equivalent) of bolts, nuts and flat washers	:	
a)	Thickness of coating <u>Minimum average coating</u>	:	
	i) Mass (gm./sq.m - Min.)	:	
	ii) Thickness (mm - Min.)	:	
b)	Electro galvanizing of spring washers	:	
	i) Mass (gm./sq.m - Min.)	:	
	ii) Thickness (mm - Min.)	:	
7.	Sampling Procedure	:	

	For Bolts & Nuts, spring washers and flat washers	:	
8.	Dimensional Particulars for G.I.Bolts, Nuts, spring washers and Flat washers	:	
	i) Shank diameter (ds) mm	:	
	ii) Width across Flats (s)	:	
	iii) Width across corners (e) ‘mm'	:	
	iv) Thickness of Bolt head (k) 'mm'	:	
	v) Length of thread (b)	:	
	vi) Thread size of Nut (m) in mm P11, dw, e, m, m1, S, etc	:	
	vii) Spring washers d1, d2, b, s etc.	:	
	viii) Flat washers: d, D, S etc	:	
9.	Whether the materials covered by the bid have been fully tested and if so whether the copies of the test certificates enclosed to the bid offer.	:	

SIGNATURE OF THE BIDDER
WITH DATE &SEAL